104.58.4. Urban Security and EFU Audit

1. The Metabolic Objective of Urban Security

Within the EFU framework, the primary function of urban security systems is to **ensure the uninterrupted operation of critical urban fluxes**—including energy, water, information, and human mobility—while minimizing entropy-generating **parasitic processes**, such as crime, vandalism, and resource misuse.

EFU Principle

Urban safety is not determined by the number of surveillance devices, but by the **self-regulatory capacity of the community and the transparency of institutional systems**.

---

2. Predictive Flux Security (PFS)

Urban security should operate as a **proactive and preventive system**, rather than a purely reactive or punitive one.

Sensor-Based Environmental Monitoring

Distributed sensor networks integrated into urban infrastructure—such as lighting poles, waste containers, and utility nodes—should monitor **environmental anomalies**, including:

• gas leaks
• sudden heat anomalies
• structural stress in buildings or infrastructure

The purpose of such systems is **infrastructure protection rather than individual surveillance**.

AI Governance and Audit Compliance

Urban security algorithms must comply with the **AI licensing standards defined in Module 104.54**.

Key requirements include:

• **ASK > 80%** — data processing must remain local to the district (edge-based processing)
• **TSI > 0.8** — decision-making logic must be transparent and auditable by the public

---

3. Measurement of the “Peace Flux”

Within the EFU framework, security is not measured by the number of arrests but by the **degree of confident public space usage**.

Nighttime Freedom of Movement Index (NFMI)

This indicator measures the extent to which residents feel safe using **interstitial urban spaces after dark**.

Audit Rule

If the movement of **women, children, or elderly residents** becomes restricted due to fear, the area receives a **Red Alert classification** and requires immediate intervention through:

• community engagement
• urban design improvements
• targeted security measures restoring the interstitium

---

4. Protection of Critical Infrastructure

Urban security institutions have a key role in protecting the **Sovereign Districts defined in Module 104.58.2**.

Physical Fallback Competence

Security personnel must be capable of manually operating the district’s **Dead-Hand systems**, including:

• water shutoff mechanisms
• emergency energy bypass systems

Communication Resilience

During large-scale disruptions, security services must be able to communicate with residents through **local edge-based communication networks**, ensuring operational continuity even if the global internet becomes unavailable.

---

5. Operational Audit Indicators

Indicator	Name	Functional Meaning

Indicator	Name	Functional Meaning
BSI	Security Stability Index	EFU loss caused by criminal activity relative to security system costs
CAI	Community Autonomy Index	Degree to which local residents can resolve conflicts without external policing
RRT	Response Time Ratio	Speed of intervention when critical flux disruptions occur (e.g., pipe rupture, power failure)

---

6. Systemic Outcome of Module 104.58.4

The implementation of this framework transforms security institutions into the **urban immune system**.

Rather than standing above the population, security structures function to **protect the metabolic homeostasis of the district** and preserve the stability of its critical fluxes.